What is Machine Learning?

Machine learning is a field of artificial intelligence in which computers learn patterns from data instead of following hand-written rules. A model is trained on examples and then used to make predictions on new data.

Supervised learning is the most common type of machine learning. In supervised learning, the training data contains inputs together with the correct answers, called labels. The model learns the mapping from inputs to labels. Example: predicting whether an email is spam from emails that people already marked as spam or not spam.

Unsupervised learning works with data that has no labels. The model looks for structure by itself, such as grouping similar customers into clusters.

Classification and Regression

Classification and regression are the two main kinds of supervised learning problems. Classification predicts a category, for example spam or not spam, or the digit shown in an image. Regression predicts a continuous number, for example the price of a house or tomorrow's temperature.

The key difference between classification and regression is the type of output: classification outputs a discrete class label, while regression outputs a real number. Classification is usually measured with accuracy, precision and recall. Regression is usually measured with mean squared error.

Random Forest

A random forest is an ensemble of many decision trees. Each tree is trained on a random sample of the training rows and considers only a random subset of the features at every split. This randomness makes the trees different from each other.

To make a prediction, every tree votes and the forest returns the majority class for classification, or the average value for regression. Averaging many different trees reduces overfitting compared to a single deep decision tree. Random forests work well on tabular data with little tuning.

Gradient Descent

Gradient descent is an optimisation algorithm used to train models. Imagine standing on a foggy hill and wanting to reach the valley. You feel the slope under your feet and take a step downhill. Repeating this brings you towards the bottom.

In machine learning the hill is the loss function, which measures how wrong the model is. The gradient tells which direction increases the loss, so we move the parameters in the opposite direction. The learning rate controls the size of each step. If the learning rate is too large the model can overshoot the minimum; if it is too small, training is very slow.